JPEDAV (2011) 32:392–393
DOI: 10.1007/s11669-011-9914-x
1547-7037 ©ASM International

Al-Ce (Aluminum-Cerium)

H. Okamoto

The Al-Ce phase diagram in [Massalski2] was redrawn from [1988Gsc]. This phase diagram was modified by [1998Oka] by introducing the information from [1996Sac] for the composition range from 65 to 100 at.% Ce.

Since then, thermodynamic modeling of this system has been attempted several times by [2001Cac], [2005Gao], [2008Kan], and [2011Jin]. Figure 1 shows the most recent work of [2011Jin] (the same author group as [2008Kan]). [2008Kan] showed a high-temperature phase Al_4Ce instead of $\beta Al_{11}Ce_3$. According to the crystal structure data in Table 1, Al_4Ce appears to be more appropriate. Further confirmation is required.

Table 1 Al-Ce crystal structure data

Phase	Composition, at.% Ce	Pearson symbol	Space group	Strukturbericht designation	Prototype
(Al)	0	*cF*4	$Fm\bar{3}m$	*A*1	Cu
$\beta Al_{11}Ce_3$	21.4	*tI*10	*I*4/*mmm*	$D1_3$	Al_4Ba
$\alpha Al_{11}Ce_3$	21.4	*oI*28	*Immm*	…	$\alpha Al_{11}La_3$
βAl_3Ce	25	…	…	…	…
αAl_3Ce	25	*hP*8	$P6_3/mmc$	$D0_{19}$	Ni_3Sn
Al_2Ce	33.3	*cF*24	$Fd\bar{3}m$	*C*15	Cu_2Mg
AlCe	50	*oC*16	*Cmc*2 or *Cmcm*	…	…
$AlCe_2$	66.7	…	…	…	…
$\beta AlCe_3$	75	*cP*4	$Pm\bar{3}m$	$L1_2$	$AuCu_3$
$\alpha AlCe_3$	75	*hP*8	$P6_3/mmc$	$D0_{19}$	Ni_3Sn
(δCe)	98.5-100	*cI*2	$Im\bar{3}m$	*A*2	W
(γCe)	100	*cF*4	$Fm\bar{3}m$	*A*1	Cu
(βCe)	100	*hP*4	$P6_3/mmc$	*A*3′	αLa

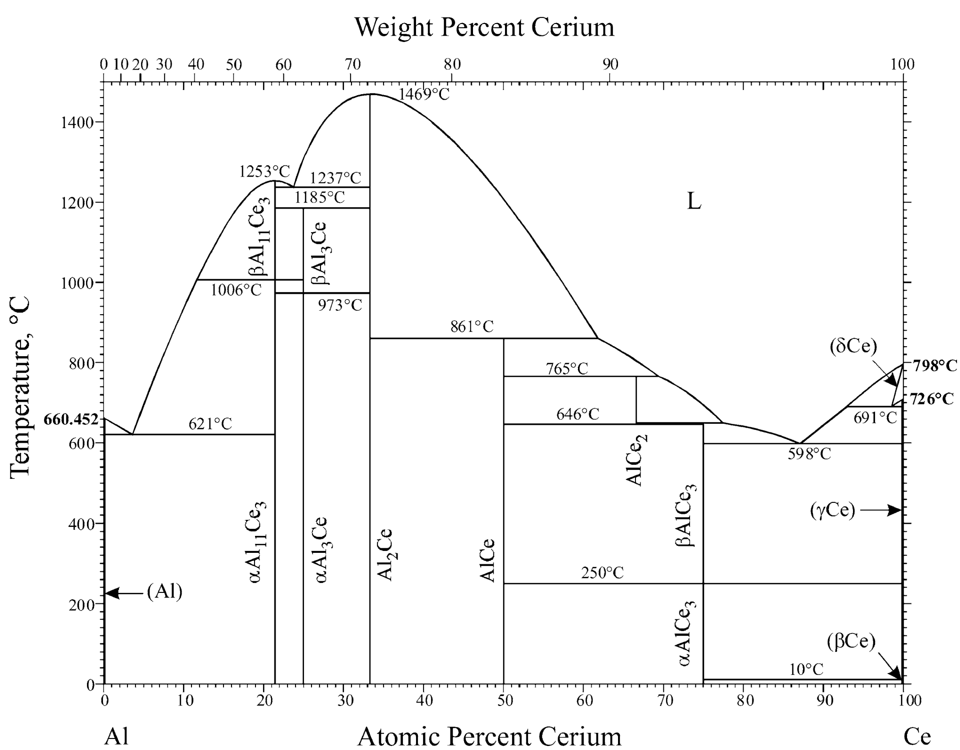

Fig. 1 Al-Ce phase diagram

References

1988Gsc: K.A. Gschneidner, Jr. and F.W. Calderwood, The Al-Ce (Aluminum-Cerium) System, *Bull. Alloy Phase Diagr.*, 1988, **9**(6), p 669-672

1996Sac: A. Saccone, A.M. Cardinale, S. Delfino, and R. Ferro, Phase Equilibria in the Rare Earth Metals (R)-rich Region of the R-Al Systems (R = La, Ce, Pr, Nd), *Z. Metallkd.*, 1996, **87**(2), p 82-87

1998Oka: H. Okamoto, Al-Ce (Aluminum-Cerium), *J. Phase Equilib.*, 1998, **19**(4), p 396

2001Cac: G. Cacciamani and R. Ferro, Thermodynamic Modeling of Some Aluminum-Rare Earth Binary Systems: Al-La, Al-Ce, and Al-Nd, *Calphad*, 2001, **25**(4), p 583-597

2005Gao: M. Gao, N. Unlu, G. Shiflet, M. Mihalkovic, and M. Windom, Reassessment of Al-Ce and Al-Nd Binary Systems Supported by Critical Experiments and First-Principles Energy Calculations, *Metall. Mater. Trans. A*, 2005, **36A**, p 3269-3279

2008Kan: Y.B. Kang, A.D. Pelton, P. Chartrand, and C.D. Fuerst, Critical Evaluation and Thermodynamic Optimization of the Al-Ce, Al-Y, Al-Sc, and Mg-Sc Binary Systems, *Calphad*, 2008, **32**, p 413-422

2011Jin: L. Jin, Y.B. Kang, P. Chartrand, and C.D. Fuerst, Thermodynamic Evaluation and Optimization of Al-La, Al-Ce, Al-Pr, Al-Nd, and Al-Sm Systems Using the Modified Quasi-chemical Model for Liquids, *Calphad*, 2011, **35**, p 30-41